

Phase 2: Smart Lead Qualification

Revised Plan v2 ADJUSTMENTS INTEGRATED NOT IMPLEMENTED

Date: June 2, 2026 | **Builds on:** Phase 1 (company_manager, COMPANY_ID)

Goal: Sales employee with lead intelligence — not a FAQ bot

v2 adjustments: contact_method · coverage_check_needed · lead_score 0–100 · ADMIN_ALERT placeholder · sales conversation memory

1. Executive Summary

Original (kept)	New (v2)
Buying intent (Nepali + English)	+ coverage_check_needed signal
Signals: budget, location, speed, urgency	+ contact_method: phone / telegram / whatsapp
Stages: new → interested → qualified → hot	+ lead_score 0–100 alongside stages
lead_manager.py persistence	+ sales memory fields on lead
Unavailable package → no list dump	+ coverage-aware sales flow
Sales employee prompt	+ ADMIN_ALERT_ENABLED placeholder

2. Revised Pipeline (16 steps)

1. sanitize + save user turn
2. extract_memory_facts (unchanged)
3. extract_lead_signals + detect_buying_intent
4. **detect contact_method (NEW)**
5. **detect coverage_check_needed (NEW)**
6. extract_lead_fields (merge memory)
7. **compute lead_score 0–100 (NEW)**
8. derive stage from score + rules
9. upsert_lead + update sales_memory
10. direct memory answer (unchanged)
11. company config check (unchanged)
12. resolve sales_mode + coverage + product match
13. compose prompt (lead_block + sales_memory_block)
14. OpenAI + save reply
15. **update sales_memory from reply (NEW)**
16. **admin_notifier placeholder if threshold met (NEW)**

Telegram → ai_employee_engine

- lead_extractor → lead_manager → leads table
- lead_score → stage (new/qualified/interested/hot)
- coverage_check_needed? → sales coverage prompt
- product match? → sales fallback (NO list dump)
- ADMIN_ALERT_ENABLED? → log placeholder
- OpenAI → reply → update sales_memory

3. Adjustment 1 — contact_method

Value	When	Stored in
telegram	Default for Telegram bot	contact_value = user_id
phone	NP mobile extracted or "call garnus"	contact_value = phone

whatsapp	User mentions WhatsApp / WA number	contact_value = WA number
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Priority: explicit mention > phone extracted > default telegram.

```
CONTACT_METHOD_PATTERNS:
  whatsapp: whatsapp, wa ma, व्हाट्सएप
  phone: phone, call, mobile, number dinchu
  telegram: telegram, telegram mai (explicit)
```

4. Adjustment 2 — coverage_check_needed

Purpose: ISP service availability validation before quoting/install.

Trigger	Example
Location + buying intent	Baluwatar ma internet jodnu cha
Location + speed ask	Pokhara ma 100 mbps chahiyo
Explicit coverage question	hamro area ma auncha?

Sales behavior: Do NOT promise install date. Acknowledge need → confirm team will verify area → collect contact.

```
coverage_check_needed  BOOLEAN
coverage_area          STRING(160)
coverage_status        pending (Phase 2 default)
```

5. Adjustment 3 — lead_score (0–100)

Factor	Points
Buying intent detected	+25
Speed/package mentioned	+15
Location/area mentioned	+15
Budget mentioned	+10
Urgency detected	+10
Phone provided	+20
WhatsApp number provided	+18
Name provided	+5
Telegram default contact	+5
Coverage check flagged	+5
Repeat buying within 24h	+5

Score → Stage mapping

Stage	Score	Rule
new	25–39	Buying intent only
interested	40–59	≥1 qualification signal
qualified	60–79	contact reachable + (location OR speed)
hot	80–100	qualified + urgency + contact_value

Stage never downgrades. Score can increase on re-upsert.

6. Adjustment 4 — Admin Notification Placeholder

```
# env.example
ADMIN_ALERT_ENABLED=false
ADMIN_TELEGRAM_CHAT_ID=
ADMIN_ALERT_MIN_SCORE=80
ADMIN_ALERT_STAGES=hot,qualified
```

admin_notifier.py — Phase 2: log-only when enabled. Phase 3: send to ADMIN_TELEGRAM_CHAT_ID.

```
logger.info("ADMIN_ALERT_PLACEHOLDER company_id=... lead_id=...
stage=... score=... contact_method=...")
```

7. Adjustment 5 — Sales Conversation Memory

Stored on **lead row** (not user_memory):

Field	Updated when
last_discussed_product	Product matched or alternative proposed
last_sales_question	User message in sales_mode
last_sales_reply	Assistant reply in sales_mode (truncate 500 chars)

```
Recent sales conversation:
- Last discussed product: 200 Mbps Internet Package
- Customer last asked: 500 mbps chahiyo Baluwatar ma
- Do not repeat full package list. Build on this context.
```

8. Complete leads Table Schema

```
leads
id, user_id, company_id
stage          - new | interested | qualified | hot
lead_score     - 0-100 (NEW)
customer_name, location, budget, requested_speed, phone
contact_method - telegram | phone | whatsapp (NEW)
contact_value  (NEW)
urgency, buying_intent
coverage_check_needed, coverage_area, coverage_status (NEW)
matched_product, alternative_product
last_discussed_product, last_sales_question, last_sales_reply (NEW)
source_message, signals_json
created_at, updated_at
```

Upsert: one active lead per (user_id, company_id), 30-day window.

9. Files Plan

Create (3)

- lead_extractor.py — intent, signals, coverage, contact_method, score, stage
- lead_manager.py — CRUD, upsert, sales_memory, lead_to_prompt
- admin_notifier.py — ADMIN_ALERT_ENABLED placeholder (NEW)

Modify (7)

- memory_db.py — Lead model (full schema)
- ai_employee_engine.py — full pipeline
- intent_engine.py — buying_intent, lead_qualification, coverage_inquiry
- prompts.py — lead_block, sales_memory_block, coverage rules
- products.py — find_best_product_match()
- env.example — LEAD_ACTIVE_DAYS, ADMIN_ALERT_*
- README_DEPLOY.md

Unchanged: main.py, openai_engine.py, company_manager.py, core memory.py

10. Combined Flow Example

User: "Baluwatar ma 500 mbps internet jodnu cha"

1. buying_intent=True
2. coverage_check_needed=True, coverage_area="Baluwatar"
3. No exact match; alternative=200 Mbps
4. contact_method=telegram (default)
5. lead_score ≈ 60 → stage=interested (until phone added)
6. sales_mode=True — NO full package list
7. Reply: acknowledge need, coverage check for Baluwatar, suggest 200 Mbps alternative, ask phone/WhatsApp
8. Update sales_memory on lead
9. If score≥80 + ADMIN_ALERT_ENABLED → log placeholder

11. Original Requirements — All Retained

#	Requirement	Status
1	Buying intent (internet jodnu cha, etc.)	✓
2	Lead signals: budget, location, speed, urgency	✓ + coverage
3	Extract name, location, budget, speed	✓ + contact_method
4	Save to lead_manager.py	✓
5	Stages: new, qualified, interested, hot	✓ + lead_score
6	Unavailable package: no list dump	✓
7	Sales employee, not FAQ bot	✓ + sales_memory

12. Testing Checklist (v2)

#	Input	Expected
1	Internet jodnu cha	score≥25, stage=new, contact=telegram
2	Baluwatar ma 100 mbps	coverage_check_needed=True
3	WhatsApp ma 9800000000	contact_method=whatsapp
4	500 mbps chahiyo	no list dump; sales_memory updated
5	name+phone+speed+urgency	score≥80, stage=hot
6	ADMIN_ALERT_ENABLED=true + hot	ADMIN_ALERT_PLACEHOLDER log

7	Second sales message	last_sales_* in prompt
8	Support hours kati ho?	no lead

13. Implementation Order (~7–8 days)

Day	Task
1	Lead model + lead_manager upsert/score/stage
2	lead_extractor — all signals + score + coverage + contact
3	products.py match + admin_notifier.py stub
4	prompts.py — lead_block, sales_memory_block, coverage
5	intent_engine + ai_employee_engine pipeline
6	Sales memory loop + Telegram tests
7	admin_export_leads.py + env/README
8	Render deploy

Estimated: ~550–650 lines across 3 new + 7 modified files

14. Out of Scope

Admin dashboard · CRM webhook · Live WhatsApp channel · Vector search · Background worker · Alembic · Coverage API (Phase 3)

15. Final Approval

- contact_method defaults to telegram — OK?
- coverage_check_needed ISP-aware (industry=isp) — OK?
- lead_score weights (\$5) — OK?
- Admin placeholder log-only — OK?
- sales_memory on lead row — OK?
- admin_export_leads.py — include or defer?

Reply **approved** to begin implementation.